Sparse Events and Complete-Tie Prefixes for Exact VUS Evaluation

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ABSTRACT

Volume Under the Surface (VUS) evaluates anomaly scores across thresholds and temporal tolerance widths. We derive an exact scalar event representation for a pinned dynamic VUS definition: thresholds without active support have analytically integrable contributions, while retained events preserve complete global score ties. With standard binary64 radix ranking and interval geometry, the evaluator takes $O(n + (L + 1)Q)$ time, where Q is maximum-buffer support. Reversible activation and incremental component unions further reduce this to $O(n + L + \alpha(K) \sum_w J_w)$, where J_w counts the active prefix through recall saturation. For one original interval, spatial runs aggregate complete score classes; an integer stopping certificate gives a run-sensitive bound without restricting tie cardinality. A fixed-point range-sum implementation has a local mass-error bound, distinct from exact real-arithmetic evaluation. A frozen three-score-vector workload covers 308 series in 17 families and 924 series–width tasks per CPU platform. At primary width 30, sparse radix passes its five-control criterion; median ratios against released fast-VUS are 20.925× and 14.538×, and against matched dense radix 2.599× and 4.791×. A separate incremental study finds that saturation fails its default-width runtime criterion in all three runs, with qualified gains at larger widths. Exactness concerns the exhaustive mathematical target. Run aggregation helps structured large ties but does not broadly improve the measured real-score runtimes. Numerical parity and practical speed are separate claims. In a directly timed 308-file evaluation, selection and output workflow, released sparse radix takes 3.187 seconds at the median, versus 11.298 seconds for matched dense radix and 63.647 seconds for fast-VUS, with identical selected-model sets across all nine passes.

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The source code, data, and/or other artifacts have been made available at https://github.com/mandu5/sparse-vus/tree/research/saturation-20261021.

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1 INTRODUCTION

Accuracy evaluation can itself be a repeated computational operation in time-series anomaly studies. Volume Under the Surface (VUS) averages range-aware ROC or precision–recall integrals across temporal tolerance widths [2, 12]. We study the evaluator after scores have been produced: given labels, scores and a maximum width, compute the same scalar for every distinct threshold and every integer width. Detector training, score generation, data ingestion and downstream scientific choice of the metric are outside this computational problem.

Existing acceleration already exploits substantial structure. Boniol et al.'s Algorithms 2 and 3 partition maximum-buffer support from unaffected regions and reuse static work [2]. Klein's fast-VUS sorts support scores and computes weighted cumulative sums [6]. Benchopt supplies a dense cumulative-sum implementation [1]. The remaining question is which threshold states and width-dependent preparations can be omitted while preserving the complete scalar integral.

Sparse VUS answers this question in stages. First, a score group with no position in maximum support changes neither weighted positive mass nor component coverage. It contributes zero PR area, but its ROC contribution must be integrated before the next retained event. Complete global tie counts preserve predictions outside support. Second, a two-distance interval lemma and standard fixed-width ranking give an exhaustive $O(n + (L + 1)Q)$ evaluator for support size Q. Third, activating only current support in score order avoids a full-support scan at each width; processing ends when both mass and component coverage permanently saturate recall. Finally, for one original interval, complete score classes can be aggregated through their contiguous temporal runs. An integer sufficient condition makes this stopping decision independent of rounded mass calculations.

These reductions account for features that distinguish the pinned target from ordinary binary AUC: positive and negative denominators depend on predicted buffer mass; expanded anomaly components merge with width; and global score ties may combine active and inactive positions. The statements are exact-arithmetic identities and operation-count bounds. Compiled floating-point realizations and their numerical contracts are evaluated separately; no bitwise identity or universal runtime advantage is inferred from a theorem.

A preliminary support-event implementation and working manuscript were publicly released as Sparse VUS 0.3.0 on 21 October 2026 [8]. This paper presents that evaluator together with its prefix and complete-tie extensions. We retain the precursor's versioned evidence and distinguish whole-evaluator improvements from incremental improvements over that precursor.

The empirical distinction matters. A frozen 308-series study evaluates three score vectors in each of 924 series–width tasks per platform; the exact radix evaluator passes its primary five-control criterion on both CPUs. A separate study measures individual score-vector calls and finds that the active-prefix extension fails its default-width primary criterion in all three runs. Larger-width improvements and slower cases are reported together. The complete-tie extension is a separate candidate with its own evidence; historical timings are not assigned to new code.

2 TARGET AND EXACT EVENT REPRESENTATION

Let $y \in \{0,1\}^n$ contain both classes and $s \in \mathbb{R}^n$ contain finite scores after binary64 conversion. Larger scores indicate greater anomaly evidence. Let $P_0 = \sum_i y_i$, and let the K maximal original anomaly intervals have inclusive endpoints $[a_k, b_k]$ in temporal order. Widths are the integers $w = 0, \dots, L$. The continuous label $a_i^{(w)}$ is one on original positives. An interval contributes $\sqrt{1 - d/w}$ outside its endpoints at integer distance $1 \le d \le \lfloor w/2 \rfloor$; contributions are summed and clipped at one. At $w = 0$ no ramp is added. Expanded intervals that overlap merge, while adjacent nonoverlapping intervals remain separate. This is the original dynamic VUS target pinned to the official `RangeAUC_volume_opt` source at revision `467bace0`, configured for exhaustive thresholds [17]. It is not shared by all software named VUS.

For a threshold t, define

$$p_i(t) = \mathbf{1}\{s_i \ge t\}, \qquad C(t) = \sum_i p_i(t), \tag{1}$$

$$A(t) = \sum_i y_i p_i(t), \qquad B(t) = \sum_i (1 - y_i) a_i^{(w)} p_i(t), \tag{2}$$

$$T(t) = A(t) + B(t), \qquad P(t) = P_0 + B(t)/2, \tag{3}$$

$$N(t) = n - P(t). \tag{4}$$

Let $H(t)$ count expanded components hit by at least one predicted positive, and let K_w count those components. The curve coordinates are

$$R(t) = \min\{T(t)/P(t), 1\}\, H(t)/K_w, \tag{5}$$

$$F(t) = (C(t) - T(t))/N(t), \qquad \pi(t) = T(t)/C(t). \tag{6}$$

Precision is used only at nonempty retained thresholds, where $C(t) > 0$; the initial origin requires no precision value. The denominator is positive: $P(t) \le (n + P_0)/2 < n$. ROC integrates trapezoids from the origin through complete distinct descending score thresholds to the explicit terminal $(1, 1)$. PR uses the right-step sum $\sum_j (R_j - R_{j-1})\pi_j$, with $R_0 = 0$. Each VUS scalar is the arithmetic mean over every width. Signed horizontal ROC increments are preserved; the derivation does not replace a dynamic denominator by a fixed one.

Let $S_w = \{i : a_i^{(w)} > 0\}$, $Q_w = |S_w|$, and $Q = Q_L$. For an active event with score v_j, precompute the complete global counts $c_j^{>} = |\{i : s_i > v_j\}|$ and $c_j^{\ge} = |\{i : s_i \ge v_j\}|$. They include positions outside S_w and outside S_L.

Lemma 1 (Inactive threshold runs). *Between consecutive active score events at a fixed width, T, B, P, N, H, R are constant. Omitted thresholds contribute a horizontal ROC rectangle and zero PR area. At an active score, its complete global numerical tie is a single transition.*

Proof. A newly admitted inactive point has zero weight and cannot hit an expanded component. Only C changes. The horizontal increments $(\Delta C)/N$ telescope at fixed height R. All positions tied with an active event enter at its global threshold, so both counts must retain them. □

If the previous retained state is (T_o, N_o, R_o, c_o), event j contributes

$$f_j^{-} = (c_j^{>} - T_o)/N_o, \tag{7}$$

$$\Delta\mathrm{ROC}_j = (c_j^{>} - c_o)R_o/N_o + (F_j - f_j^{-})(R_j + R_o)/2, \tag{8}$$

$$\Delta\mathrm{PR}_j = (R_j - R_o)T_j/c_j^{\ge}. \tag{9}$$

The initial state is $(0, n - P_0, 0, 0)$, and afterwards $c_o = c_j^{\ge}$. Connecting retained endpoints without the intervening rectangle is incorrect. For example, $y = (1, 0, 1, 0)$, $s = (3, 2, 1, 0)$ and $L = 0$ give ROC area 0.625; connecting support endpoints alone gives 0.8125.

Lemma 2 (Two-distance geometry). *The weight of a nonoriginal point is determined by its distances d_1, d_2 to the nearest two distinct original intervals. It is zero if $d_1 > \lfloor w/2 \rfloor$, one if $d_2 \le \lfloor w/2 \rfloor$, and $\sqrt{1 - d_1/w}$ otherwise.*

Proof. Every active ramp has weight at least $1/\sqrt{2}$. Two contributions therefore clip to one. If exactly one interval contributes, its distance determines the weight. Distances to ordered intervals are monotone moving away on each side, so two neighbors on each side contain the nearest two intervals. □

Theorem 1 (Support-event evaluation). *In a unit-cost model with score comparisons and metric arithmetic, including square roots, both exhaustive scalar integrals can be computed in $O(n \log(Q+1) + (L+1)Q)$ time and $O(n + Q)$ auxiliary words.*

Proof. Extract original intervals and maximum support in $O(n)$ time. Sort the Q support scores and group equal values. Binary-search all n scores among these groups to obtain complete global counts $c^{>}, c^{\ge}$ without sorting off-support scores, at cost $O(n \log(Q + 1))$. The ordered-interval geometry can be prepared within this bound. At each width, compute support weights, component membership, grouped masses and first component-hit events in $O(Q + K)$ time. A scan of the at most Q event groups integrates the preceding rectangles, complete tied transitions and terminal ROC tail. The inactive-run lemma makes this identical to the exhaustive target in exact metric arithmetic. Stream the width mean using input-, support- and interval-sized arrays. Since $K \le Q \le n$, the stated bounds follow. □

Proposition 1 (Fixed-binary64 preparation). *For finite binary64 scores, complete global tie counts, the score ordering of maximum support, and its two-distance geometry can be prepared in $O(n + Q + K)$ time and $O(n + Q)$ auxiliary words. Discrete work uses a word-RAM with input indices, counts, widths, and intermediate index arithmetic fitting words. Metric arithmetic, including square roots, has unit cost.*

Proof. Normalize signed zeros. Complement a negative score's bit pattern and flip the sign bit of a nonnegative score, then complement the result to obtain a descending numerical key. Eight stable byte-counting passes order these 64-bit keys in $O(n)$ time. Scan complete ties and retain their starting and ending global ranks. Filter the permutation to maximum-support positions. A temporal scan with a monotone interval pointer computes nearest distances and owners in $O(Q + K)$ time. All arrays are input- or support-sized. □

Radix sorting and floating-point key transforms are established techniques [3, 5]. This assumption does not give linear sorting of arbitrary real scores or a bit-complexity bound for growing precision. Conversion defines numerical ties, including collisions between distinct large integer inputs.

The same support-event width kernel combined with this preparation gives $O(n + (L + 1)Q)$ time. This is the basic Sparse VUS evaluator; its improvement over a dense rank sweep depends on Q, rather than recall saturation. The released PR-only implementation retains $L + 1$ scalar width results, adding $O(L)$ words; streaming removes this storage. Input-sized ranking arrays remain necessary in this construction, so the memory is not support-only.

3 PRIOR ALGORITHMS AND COMPARISON SCOPE

Boniol et al.'s optimized VUS algorithms retain threshold loops within widths and support-region summation [2]. The official source additionally creates full-length prediction and label arrays per threshold. Klein removes repeated threshold-wise summation through support cumsums, but still constructs maximum-support and full-length coverage arrays at each width [6]. Benchopt cumsums over all positions and computes component-hit counts through segment maxima [1]. Table 1 distinguishes these mechanisms from the new reductions. Its costs describe inspected operations, not cross-library runtime predictions.

Sorted sweeps, complete ties and trapezoids are established ROC techniques [4]. scikit-learn 1.0.2 explicitly truncates PR at full recall after forming cumulative counts [13]. Therefore full-recall stopping and zero-area PR suffixes are not new principles. Our comparison includes last-original-positive stopping, which already ensures saturation. The specific questions are whether buffer mass permits an earlier stop and whether preparation avoids paying for the discarded suffix. Reversible list updates [7], union-find [16], radix ordering [3, 5] and exact accumulators [11] are established tools used in the construction.

Sylligardos reports preliminary CPU/GPU VUS acceleration [14]. The public FF-VUS code attributed to him in aVUSi [15] uses a different ramp/overlap contract; its wrapper rounds scores, and Torch can interpolate component existence across widths when memory is insufficient. This is relevant acceleration prior art, but its native runtime is not an exact-target comparison. No result here establishes superiority over an unmeasured GPU evaluator. Source pins and minimal semantic witnesses are retained with the artifact; a package name alone does not establish equality.

Once: rank complete global ties; prepare interval distances and owners; schedule activations/unions; record reverse-deletion neighbors.

For each width: restore newly active positions and merge overlapping intervals; start mass, hit count and areas at zero.

For each complete active score tie: update weighted mass and component hits; integrate its preceding negative rectangle and tied transition.

Stop: if $A + B/2 \geq P_0$ and $H = K_w$, append ROC tail $1 - F$; the remaining PR area is zero. Stream the width means.

Figure 1: Position-prefix evaluator. No width clears or scans a support-sized or component-sized array before traversal.

4 PREFIX EVALUATION WITHOUT FULL-SUPPORT SETUP

4.1 Exact evaluation of active score prefixes

At a fixed width, let A count admitted original-positive positions and let B denote their admitted non-original buffer mass. Then $T = A + B$ and $P = P_0 + B/2$. We call a complete score threshold *saturated* when

$$A + B/2 \geq P_0 \qquad \text{and} \qquad H = K_w. \tag{10}$$

The condition includes the complete global numerical tie, even when only some positions in that tie belong to the width's active support.

Lemma 3 (Permanent saturation). *After a threshold satisfying* (10)*, every subsequent recall coordinate equals one. The remaining right-step PR contribution is zero, and the remaining trapezoidal ROC contribution, including the explicit terminal* $(1, 1)$ *transition, is* $1 - F$*, where* F *is the current FPR.*

Proof. Threshold relaxation cannot decrease A, B, or the number of hit expanded components. Thus both conditions persist. The first is equivalent to $(A + B)/(P_0 + B/2) \geq 1$; the second makes the existence factor one. Every later PR recall increment vanishes. Every later ROC trapezoid has height one, so its signed horizontal increments telescope from the current F to the terminal abscissa one. The proof does not assume a fixed negative denominator. □

Stopping at recall one is a standard flat-curve observation, not a claimed new integration principle. In particular, admitting the last original positive gives $A = P_0$ and hits every expanded component, so it is already a valid cutoff. The VUS-specific condition can stop earlier: buffer mass can compensate for original positives that have not yet been admitted. Both last-original-positive stopping and disabled stopping are explicit controls.

A cutoff added after constructing every width's full support histogram would still pay $\Theta(Q)$ per width. To avoid that work, retain support positions in descending score order, and activate position i at radius $d_1(i)$; original positives have activation radius zero. The score-ordered list is represented by integer next pointers. Bucket-sort activations by their integer radius. Starting from the full list, delete positions in reverse activation order and record the two neighbors at each deletion. Reversing these deletions inserts every position in constant time. Tied activation radii use the same fixed

Construction	Threshold and width work	Semantic/comparison qualification
Boniol Algorithms 2/3	Per-width threshold loop; Q-support sums per threshold	Static-support reuse and cached predictions are prior techniques; the released target uses $\geq$ ties.
Klein fast-VUS	Per-width support cumsum, full-series coverage and threshold arrays	PR only; use $H = n$, not the sampled default $H = 250$. Explicit arrays cost $\Omega(Q + n + H)$ per width.
Benchopt	Per-width full-series cumsum and threshold arrays	Computes both ROC and PR; $H = n$ covers exhaustive ties. Explicit arrays cost $\Omega(n + H)$ per width.
FF-VUS	Batched width/threshold matrices and support masking	Native linear/max ramps differ from the pinned target; public wrapper rounds scores; Torch has an existence approximation fallback.
Sparse support events	$O(n + (L + 1)Q)$ with fixed-key preparation	Removes inactive thresholds while retaining global counts and exact ROC rectangles; no prefix stop is required.
Active positions	$O(n + L + \alpha(K)\sum_w J_w)$	Complete terminating active ties; arbitrary components; worst-case $J_w = Q_w$.
Complete-tie runs	$O(n + L + \sum_w \sum_{g \in \mathcal{G}_w} R_g)$	One original interval; charges all stored runs of every visited class; integer stopping certificate.

Table 1: Algorithmic comparison. H is the requested threshold count; Q is maximum support; J_w counts visited active positions; R_g counts temporal runs of score class g. External array-cost lower bounds omit additional ramp, sorting and search work. Batched FF-VUS has no single sequential per-width loop. Our bounds use the stated arithmetic model.

order in both directions. Consequently traversing the current list visits only positions with nonzero weight at that width. A complete retained event still uses its precomputed global $c^{>}, c^{\geq}$; points omitted from the current list may belong to that same global tie.

For geometry, preprocessing also records one nearest original interval for each support position. At radius r, neighboring original intervals with endpoints b_k, a_{k+1} merge precisely when $2r \geq a_{k+1} - b_k$. Bucket these edges by $\lceil (a_{k+1} - b_k)/2 \rceil$ and maintain their components with union-find. Mere adjacency of expanded intervals does not merge them. An active point lies in its nearest interval's expansion. Any other expansion containing the point overlaps that expansion and belongs to the same component; hence the nearest interval's union-find root identifies the point's component. A width-stamped root array counts first hits without clearing all components at each width. Unions occur before a width's threshold scan, so roots are fixed throughout that scan.

Theorem 2 (Active-prefix bound). *Use the fixed-binary64 word-RAM and unit-cost metric-arithmetic assumptions of Proposition 1. Let J_w be the number of active support positions visited through the complete first saturated score tie at width w, and let K count original intervals. Both scalar integrals can be computed in*

$$O\left(n + L + \alpha(K) \sum_{w=0}^{L} J_w\right)$$

time and $O(n+Q)$ auxiliary words, where α denotes the inverse Ackermann factor for union-find. A diagnostic vector of per-width work counts uses an additional $O(L)$ words. The values equal the target integrals in exact metric arithmetic.

Proof. Fixed-width radix ranking, complete global tie counts, the nearest-interval scan, and extraction of the ordered support take $O(n + Q + K)$ time. All activation distances and finite merge radii are at most $n-1$, so bucket arrays are capped at $\min(\lfloor L/2 \rfloor, n-1)+1$, even if L is much larger than n. Counting-sort schedules and reverse deletions take $O(n+Q+K)$ time and space. Each position is activated once, and there are at most $K-1$ unions. Union by size with path compression takes amortized $O(\alpha(K))$ per operation. Every visited position performs one find; event arithmetic and first-hit accounting take constant additional time. No width requires an additional Q- or K-sized scan. Streaming the means requires $L + 1$ width iterations. Since at width zero saturation requires all original positives, $J_0 = P_0 \geq K$, and the union cost is absorbed by $\alpha(K)\sum_w J_w$. Active-event elimination uses the same negative-run rectangles as before, and the lemma supplies the exact omitted suffix. All scratch state is input-, support-, or interval-sized. □

The bound is adaptive, not a worst-case improvement for every score vector: J_w may equal the full active support size. Union-find and linked-list access also have costs that a work-count reduction does not measure. Nevertheless, a separating family exists even against the same active-list algorithm stopped at the last original positive. For $L \geq 2$, take $n = 4L+3$, with singleton originals at zero-based positions $L + 1$ and $3L + 1$. Rank first the first original, then its two immediate neighbors, then one immediate neighbor of the second original. Rank the second original last; put all remaining scores strictly between the fourth and last scores. At every width $w \geq 2$, those three neighbors have weight at least $1/\sqrt{2}$, so the first four active positions satisfy

$$A + B/2 \geq 1 + \frac{3}{2\sqrt{2}} > 2 = P_0$$

and hit both expanded components. Thus $J_w = 4$, while at widths zero and one $J_w = 2$. The last-original-positive control must traverse $Q_w = 4\lfloor w/2 \rfloor + 2$ active positions at every width, giving $\Theta(L^2)$ traversal rather than $\Theta(L)$. With $K = 2$ and $n = \Theta(L)$, the full adaptive algorithm takes $O(n+L)$ on this family. This is a structural operation-count separation, not an empirical runtime claim.

The implementation uses finite-precision arithmetic and can differ from a full scan through summation order and suffix telescoping. The theorem is an exact-arithmetic identity; measured numerical parity, floating-point limits, and runtime comparisons must be reported separately.

4.2 A uniform one-interval guarantee

Every active original contributes one to $A + B/2$. Every active non-original contributes at least $1/(2\sqrt{2})$, because its half-width ramp is at least $1/\sqrt{2}$. After any q admitted active positions, mass is therefore at least $q/(2\sqrt{2})$, even inside a tie. Only complete ties are admissible thresholds.

Proposition 2 (Distinct active scores in one interval). *If $K = 1$ and scores in S_L are pairwise distinct after binary64 conversion, then $J_w \le \min\{Q_w, \lceil 2\sqrt{2}P_0 \rceil\}$ for every width and score ordering. The position-prefix evaluator takes $O(n + LP_0)$ time. A singleton has $J_w \le 3$ and $O(n + L)$ time.*

Proof. The first active position hits the only expanded component. The preceding mass inequality ensures saturation by $\lceil 2\sqrt{2}P_0 \rceil$ positions; if support is smaller, admitting it all includes every original and saturates. Distinct active scores permit stopping after each position. Union-find is constant work for one interval, and summing over widths gives the bound; $P_0 \le n$ absorbs the additional width-zero term. □

The guarantee needs no score-quality assumption. It does require distinct *active* scores: exclusively off-support ties do not enlarge active events. For an all-tied singleton, a complete event contains Q_w active positions and the position-wise evaluator can perform $\Theta(L^2)$ total visits when $n = \Theta(L)$. This is a limitation of that evaluator, not a problem lower bound. For multiple components, a low-scoring component can also force a long coverage prefix after enough mass has accumulated. The next construction addresses tied-position enumeration for one interval.

4.3 Complete-tie aggregation for one original interval

This subsection describes a separate candidate evaluator. It does not alter the frozen position-by-position evaluator or its recorded measurements. Assume there is exactly one original interval $[a, b]$, with $P_0 = b - a + 1$. Its maximum support S_L is a single interval. For each validated numerical score class g in S_L, store its maximal contiguous spatial runs, and let R_g be their number. These runs are defined in the maximum support, not separately at each width. A class becomes active when the width radius first reaches the minimum distance of one of its positions from $[a, b]$. Classes, rather than individual positions, can therefore use the same reverse-deletion insertion schedule, in descending score order. Global prediction counts still include every position in a complete global tie, including positions outside S_L.

Lemma 4 (A ramp run is one square-root range sum). *Let $w \ge 2$, and let a nonempty run fragment on one side of $[a, b]$ have active distances $1 \le d_- \le d_+ \le \lfloor w/2 \rfloor$. Writing $U(t) = \sum_{j=1}^{t} \sqrt{j}$, with $U(0) = 0$, its buffer mass is*

$$\sum_{d=d_-}^{d_+} \sqrt{1 - d/w} = \frac{U(w - d_-) - U(w - d_+ - 1)}{\sqrt{w}}.$$

Every stored run has at most two such fragments and one original fragment. Thus its exact mass and admitted-position counts require constant work given exact range sums. At widths zero and one, only the original fragment contributes.

Proof. The distance to a single interval changes by one at consecutive positions on either side. The identity follows from $\sqrt{1 - d/w} = \sqrt{w - d}/\sqrt{w}$ and reversing the summation order. Intersections with the left ramp, original interval, and right ramp are interval intersections and require constant work. □

Lemma 5 (Integer saturation certificate). *At the end of a complete active score class, let A count admitted original positions and let D count admitted non-original active positions. The test*

$$\lfloor D/3 \rfloor \ge P_0 - A$$

certifies true mass-and-coverage saturation. It uses only integer counts, and the test $D \,//\, 3 \ge P_0 - A$ avoids multiplication overflow. It may stop later than the real-valued mass test. Nevertheless, the number of complete active score classes visited at any width is at most $3P_0$, without any tie-size assumption.

Proof. Every active non-original weight is at least $1/\sqrt{2} > 2/3$, hence $B/2 \ge D/(2\sqrt{2}) \ge D/3$. The certificate implies $A + B/2 \ge P_0$. There is one expanded component, hit by the first active class. If $q = A + D \ge 3P_0$ positions have been admitted, then $D = q - A \ge 3P_0 - 3A$, so the certificate holds. If fewer than $3P_0$ active positions exist, admitting them all gives $A = P_0$ and also certifies saturation. Each visited active class contains at least one new position. Consequently, after at most $3P_0$ complete classes the certificate holds, regardless of the cardinality of those classes. □

Let $\mathcal{G}_w$ be the classes visited through the first complete class satisfying the integer certificate. The bound below charges all R_g maximum-support runs of each visited class, including those inactive at that particular width.

Theorem 3 (Tie-run work bound). *Under the fixed-key and unit-cost exact metric-arithmetic model, the ideal run-aggregated evaluator computes both scalar integrals in*

$$O\left(n + L + \sum_{w=0}^{L} \sum_{g \in \mathcal{G}_w} R_g\right)$$

time and $O(n + L)$ auxiliary words. For $R_{\max} = \max_g R_g$, its time is $O(n + L + (L + 1)P_0 R_{\max})$. For a singleton original and bounded $R_{\max}$, it is $O(n + L)$ even with arbitrary active tie cardinalities. If all support scores are tied, it is $O(n + L)$ for every one-interval length P_0.

Proof. Radix ranking, global tie counts, temporal run extraction, class activation distances, and the offline class insertion schedule take $O(n)$ time and space. Each class is inserted once. The $L + 1$ square-root prefixes take $O(L)$ time and space in exact metric arithmetic. A visited class costs $O(R_g)$ to compute its mass and counts. The one-component existence factor requires constant work. After the complete class, the standard negative-run rectangle, event trapezoid, and right-step PR formulas use these exact event aggregates. Lemma 5 makes the omitted suffix valid by permanent saturation. Summing gives the first bound; $|\mathcal{G}_w| \le 3P_0$ gives the second. For an all-tied support, there is one class containing one run, visited once per width. □

The hypothesis is on score-class fragmentation, not the number of distinct scores. An alternating high/low score pattern can give $R_g = \Theta(n)$, so this theorem is not a universal linear-time result for arbitrary ties. The extra $O(L)$ prefix storage is present even without diagnostics. It differs from the frozen evaluator's $O(n+Q)$ auxiliary-space bound.

A cancellation-free finite-precision range-sum implementation. Fixed-point superaccumulators are established summation techniques [11]. Subtracting two ordinary floating-point square-root prefixes can lose relative accuracy on short ranges. Instead, let q_j be the binary64 evaluation of $\sqrt{j}$ and accumulate the integers $2^{52}q_j$ exactly. For $1 \le j \le L \le 2^{63}-1$, $1 \le q_j < 2^{32}$, and every $2^{52}q_j$ is an integer below 2^{84}. Every prefix is therefore below 2^{147} and fits in three unsigned 64-bit limbs. Carry and borrow operations give exact integer prefixes and exact range subtraction in a constant number of word operations. If a nonnegative range difference has limbs (h, m, ℓ), high to low, convert it by

$$\mathrm{fl}\big(2^{76}\,\mathrm{fl}(h) + 2^{12}\,\mathrm{fl}(m) + 2^{-52}\,\mathrm{fl}(\ell)\big)\,,$$

then divide by the evaluated $\sqrt{w}$. All additions are nonnegative; there is no subtraction of rounded large prefixes. This preserves the stated operation and storage bounds, up to a constant of three words per prefix. It exactly sums the *rounded* square-root table, not symbolic square roots.

Proposition 3 (Local range-mass error). *Assume binary64 round-to-nearest operations, the usual relative error bound for integer conversion and square root, no reassociation, and exact limb arithmetic. Put $u = 2^{-53}$ and $\gamma_k = ku/(1-ku)$. The preceding implementation's nonempty ramp-fragment mass $\widehat{B}$ satisfies*

$$\left|\widehat{B}/B - 1\right| \le \gamma_8.$$

If every table index and width is at most 2^{53}, γ_6 suffices. These bounds do not grow with the prefix index or the fragment length. Summing t nonempty fragment masses using ordinary positive additions gives a relative bound γ_{t+7}, provided $(t+7)u < 1$.

Proof. Integer conversion followed by square root places each $q_j/\sqrt{j}$ between $(1-u)^2$ and $(1+u)^2$. Exact nonnegative summation preserves these bounds. Conversion of each limb and two nonnegative additions add at most three rounding factors; power-of-two scaling is exact and in range. Division adds one factor, and the evaluated denominator contributes at most two factors. Thus the final ratio lies between $(1-u)^6/(1+u)^2$ and $(1+u)^6/(1-u)^2$, contained in $[1-\gamma_8, 1+\gamma_8]$. Exact integer conversion removes one factor from both numerator and denominator, yielding γ_6. The standard positive-sum product bound combines γ_8 with γ_{t-1} to give γ_{t+7}. □

This is a local mass-error bound. It is not an end-to-end error certificate for the integrated scalar metrics, nor a claim of bitwise agreement with the frozen $\sqrt{1-d/w}$ evaluation. The integer stopping certificate is independent of these mass-rounding errors, but metric arithmetic and summation still use finite precision. The work bound does not imply runtime dominance; the experiments examine both compressible and fragmented ties.

4.4 Implementation and arithmetic scope

The support-event and position-prefix implementations expose both PR-only and paired ROC–PR calls. Validation, score conversion and all preparation belong to the public call. The PR route omits ROC arithmetic; diagnostics that return per-width work counts are outside metric timing. Position-prefix ablations use identical preparation but either disable stopping or stop at the last original positive. The complete-tie candidate uses a separate integer sufficient condition and its own measured scope.

All integer widths are evaluated. Consecutive widths with the same radius can have different ramp weights: for $y = (0, 1, 0)$ and $s = (2, 1, 0)$, width-two PR is approximately 0.77705 and width-three PR approximately 0.85505. Reusing an area merely because $\lfloor w/2 \rfloor$ is unchanged would alter the target.

The position-prefix mass test uses floating point in the implementation; its real-arithmetic proof is not a directed-rounding certificate. The complete-tie integer certificate avoids that particular stopping uncertainty, but its local range-mass bound is not an end-to-end integrated-scalar error bound. Both implementations can differ from a full scan through rounded roots, accumulation order and suffix telescoping. Numerical tolerances and independent-reference checks are reported separately from the exact identities.

5 EXPERIMENTAL DESIGN AND EVIDENCE SEPARATION

We distinguish two frozen studies with different task units and different questions. Neither trains or queries a detector during evaluation timing. The new complete-tie candidate has a separate evidence record; it does not inherit either study's timings.

Whole-evaluator study. The v3 policy holdout uses 308 series from 17 represented families of the previously studied corpus, with the released Chronos-Bolt base, Time-MoE 200M and TimesFM 1.0 200M score vectors at context length 64 [9, 19]. It is a held-out policy evaluation within previously studied source families, not a new-domain dataset. Each task is one *series–maximum-width* pair: evaluate all three frozen score vectors and return the best-PR model set under the frozen numerical tie convention. There are $308 \times 3 = 924$ tasks per platform for maximum widths 30, 100 and 300, with 308 primary tasks at width 30. This is distinct from 924 individual score-vector calls. It measures metric evaluation and a fixed three-way comparison, not end-to-end model selection or detector quality.

The exact sparse-radix evaluator is compared with five frozen controls: comparison-based sparse PR, direct dense PR, nearest-distance dense ROC–PR, direct dense radix PR, and nearest-distance dense radix PR. The matched radix controls share fixed-key ranking, and the nearest control shares interval geometry; their full-series width traversal isolates support-event compression from those ingredients. Actual pinned Klein PR and Benchopt ROC–PR are additional descriptive external comparisons, outside the five-control decision. All use exhaustive thresholds and widths. Paired outputs are explicitly marked in the tables.

Complete-task timings include validation and preparation for the three score vectors. Three warmed repetitions run serially in seeded randomized method order with one compute thread. Data I/O, imports and declared JIT warmup are excluded. The

primary decision requires numerical agreement and lower 95% family-cluster bootstrap bounds above one for both geometric and aggregate ratios against every frozen control. The artifact retains the exact inputs, sources, raw timings and decisions. The absolute numerical parity tolerance is 10^{-11}; selected-model ties use absolute tolerance 10^{-10}. Apple M1 Pro and hosted Linux results are separate runs; software versions, process snapshots and source hashes are recorded per run. They are not pooled, and environment differences are not experimentally separated.

Incremental position-prefix study. Here a case is one *series–score-vector–maximum-width* PR call. The candidate is compared with eight controls: full active traversal, last-positive stopping, released sparse radix, nearest dense radix, Klein fast-VUS, comparison-based sparse, comparison-based direct dense, and direct dense radix. All nine methods compute PR only. Klein receives n ranked thresholds; repeated ties have zero recall increments.

The development set has 42 previously studied series and three released score vectors per series. The external set was selected from mTSBench metadata before generating scores or observing metric outcomes [18]: exclude the 23 earlier source-family names with recorded alias normalization, restrict test lengths to 100,000, and take the lexicographically first two eligible series per family. This yields 12 series from nine source families and 457,171 rows. Source-family disjointness does not establish independent real-world domains; synthetic and reused data are identified in the manifest. Four fixed label-free CPU procedures generate score vectors: Isolation Forest, standardized PCA reconstruction, reference-set nearest-neighbor distance, and robust marginal deviation. These are evaluation inputs, not detector leaderboard claims; their generation is outside timing.

Widths 30, 100 and 300 give 378 development cases and 144 external cases per host. Each primary external width has 48 score vectors from 12 series. Three randomized warmed serial repetitions include all public-call work; imports, I/O, JIT warmup and diagnostic profiling are excluded. Libraries and Numba use one compute thread. The primary criterion requires maximum absolute PR discrepancy at most 10^{-11} and both 95% family-cluster bootstrap lower bounds above one against each of the eight controls at width 30. Its 2,000 family resamples quantify observed family/case variation, not three-repeat timing uncertainty or a population of all possible domains. The Mac uses M1 Pro/Python 3.12.4; the hosted Ubuntu 24.04 runner reports EPYC 9V74/Python 3.12.15. Both use NumPy 1.26.4 and Numba 0.59.1 for this study. These versions are not assumed for the earlier whole-evaluator study.

For all reported ratios the numerator is control time and the denominator is candidate time. The two preceding studies' aggregate ratios divide sums of per-case median times; they are not directly timed application batches. Section 6.2 separately times a complete stored-score collection operation, including input reads and output serialization. No primary decision is replaced by a favorable secondary width or a post-measurement comparator.

6 RESULTS

6.1 Frozen v3 policy holdout: exact radix evaluation

Each platform completed all 924 tasks on the same 308 series in 17 families, with $L = 30$ primary, $L = 100$ secondary, and $L = 300$ a stress regime. Each task evaluates the three frozen model score vectors and returns the best-PR model set, retaining numerical ties. Times include whole-task preparation and use the median of three repetitions. The maximum absolute PR discrepancy from the comparison-sort sparse PR reference was 1.554×10^{-15} on Mac and 1.332×10^{-15} on Linux. These are numerical agreement checks for the pinned target.

The frozen exact-radix decision at $L = 30$ is Mac: **PASS**; Linux: **PASS**. It requires both the geometric-mean and aggregate-ratio lower confidence bounds to exceed one against every one of the five frozen controls, together with correctness. Klein and Benchopt comparisons are descriptive and are outside this decision. Table 2 retains every primary control and both external implementations.

All ratios are comparator time divided by exact sparse-radix PR time; values above one favor radix PR. Brackets contain audited 95% family-cluster bootstrap intervals (2000 resamples). The aggregate ratio divides sums of per-case median times; it is not an observed batch wall time. Intervals describe variation across families, not uncertainty in three timing repetitions. F/S/T counts cases where radix PR is faster/slower/tied. Decisions use unrounded bounds.

Secondary and stress regimes. Table 3 gives both matched PR-only radix dense controls at $L = 100$ and $L = 300$. These controls share fixed-width score ordering with sparse radix and retain full dense width sweeps. Their ratios assess the support-event reduction after matching the ranking implementation. The full bound reports retain every other comparator, family effect, and slower case at each width.

Post-protocol descriptive PR-only oracle. For each case, the oracle selects the fastest of comparison-based sparse PR, direct dense PR, Klein PR, direct dense radix PR, and nearest-interval dense radix PR. Paired ROC+PR methods and adaptive racers are excluded from this PR-only exhaustive set. Table 4 and Figure 2 retain every case and loss. No confidence interval is assigned to this retrospective minimum, and it changes no frozen decision.

Separate racing decisions and scope. The frozen comparison-order racing decision at $L = 30$ is Mac **FAIL**, Linux **FAIL**; the separate radix-order racing decision is Mac **FAIL**, Linux **FAIL**. The first requires both interval lower bounds above one against comparison-based sparse PR, direct dense PR, and exact radix PR; the second additionally requires this against both matched dense radix PR controls. These decisions include preparation, adaptive queries, and exact completion in each measured task. Racing outcomes are separate from the exact-radix decision, and failed criteria remain reported. The v2 development runs remain historical development evidence and are not pooled with this holdout. Platforms are summarized separately: no cross-run timing pooling is performed. The Mac was not isolated: its starting snapshot

Table 2: Primary $L = 30$: all 308 cases per platform. † marks the five frozen decision controls. R+P methods compute both ROC and PR; the others compute PR only. Dense direct and dense nearest use comparison sorting unless labelled radix.

Platform	Comparator	Median	Geomean [CI]	Aggregate [CI]	F/S/T
Mac	Sparse comparison†	1.774	1.620 [1.421, 1.920]	1.960 [1.511, 2.323]	271/37/0
Mac	Dense direct†	3.331	3.160 [2.772, 3.506]	4.255 [3.349, 5.112]	306/2/0
Mac	Dense nearest (R+P)†	5.050	4.840 [4.356, 5.368]	6.140 [4.974, 7.181]	308/0/0
Mac	Dense direct radix†	2.677	2.617 [2.318, 2.870]	3.595 [2.752, 4.514]	303/5/0
Mac	Dense nearest radix†	2.599	2.519 [2.297, 2.726]	3.298 [2.624, 3.980]	307/1/0
Mac	Klein fast-VUS	20.925	22.360 [19.764, 24.238]	19.450 [17.910, 20.623]	308/0/0
Mac	Benchopt (R+P)	35.420	37.572 [34.754, 41.101]	37.990 [34.728, 42.965]	308/0/0
Linux	Sparse comparison†	1.987	1.916 [1.839, 2.013]	1.955 [1.745, 2.095]	304/4/0
Linux	Dense direct†	4.894	4.161 [3.170, 5.008]	5.385 [4.241, 6.191]	308/0/0
Linux	Dense nearest (R+P)†	6.917	5.890 [4.639, 6.923]	7.214 [5.688, 8.187]	308/0/0
Linux	Dense direct radix†	4.419	3.756 [2.904, 4.486]	4.708 [3.794, 5.276]	308/0/0
Linux	Dense nearest radix†	4.791	4.047 [3.142, 4.821]	5.010 [3.933, 5.643]	306/2/0
Linux	Klein fast-VUS	14.538	13.814 [11.639, 15.930]	11.232 [10.034, 12.514]	308/0/0
Linux	Benchopt (R+P)	21.493	20.973 [18.481, 23.617]	22.991 [19.193, 30.937]	308/0/0

Table 3: Secondary and stress widths: all 308 cases per row, matched PR-only controls. Ratio direction and intervals follow Table 2.

Platform	L	Comparator	Median	Geomean [CI]	Aggregate [CI]	F/S/T
Mac	100	Dense direct radix	5.110	4.834 [3.728, 5.802]	6.874 [4.488, 8.943]	307/1/0
Mac	100	Dense nearest radix	4.637	4.551 [3.602, 5.370]	6.096 [4.117, 7.577]	305/3/0
Mac	300	Dense direct radix	6.916	6.534 [4.259, 8.903]	8.850 [5.163, 12.681]	308/0/0
Mac	300	Dense nearest radix	6.294	6.000 [3.902, 8.163]	7.892 [4.670, 11.080]	300/8/0
Linux	100	Dense direct radix	7.465	6.666 [4.438, 8.952]	8.220 [5.435, 10.714]	308/0/0
Linux	100	Dense nearest radix	7.919	6.975 [4.574, 9.447]	8.543 [5.360, 11.345]	302/6/0
Linux	300	Dense direct radix	7.612	7.360 [4.333, 10.986]	8.808 [5.506, 13.079]	308/0/0
Linux	300	Dense nearest radix	8.103	7.426 [4.155, 11.417]	8.943 [5.047, 13.692]	295/13/0

shows an unrelated Python worker near 100 percent CPU, alongside normal desktop activity. Snapshots are not a continuous load trace. Agent test, build, and report execution stopped during timing; lightweight status reads, completed remote-artifact transfers, and document edits continued. No unrelated process was stopped by the experiment. The Linux confirmation used a separate serial GitHub Actions job; hosted hardware and host background load were outside our control. This policy holdout uses previously studied corpus families; it is not an external-domain dataset. The results address the stated workload and tested implementations, without establishing universal superiority.

Reproducibility receipt (SHA-256 prefixes): Mac report `c72f2df666c4`, audit `a9d44deb51fa`; Linux report `46c6fba361e4`, audit `577dab610dd4`. The accompanying JSON receipt records full report, measured-source, and generator hashes.

Absolute workload costs. For the 308 primary three-score tasks, sums of per-task median times are Mac: 3.094, 10.204, and 60.186 seconds (sparse radix, matched nearest dense radix, and fast-VUS, respectively). Linux: 2.318, 11.610, and 26.032 seconds (sparse radix, matched nearest dense radix, and fast-VUS, respectively). At $L =$

Table 4: Descriptive PR-only oracle / exact sparse-radix ratios; no confidence intervals. Minimum retains the slowest relative radix case.

Platform	L	Median	Geomean	Aggregate	F/S/T	Minimum
Mac	30	1.709	1.582	1.924	268/40/0	0.531
Mac	100	1.557	1.500	1.734	286/22/0	0.618
Mac	300	1.278	1.293	1.375	285/23/0	0.610
Linux	30	1.986	1.879	1.928	302/6/0	0.807
Linux	100	1.715	1.655	1.612	301/7/0	0.815
Linux	300	1.275	1.309	1.235	294/14/0	0.770

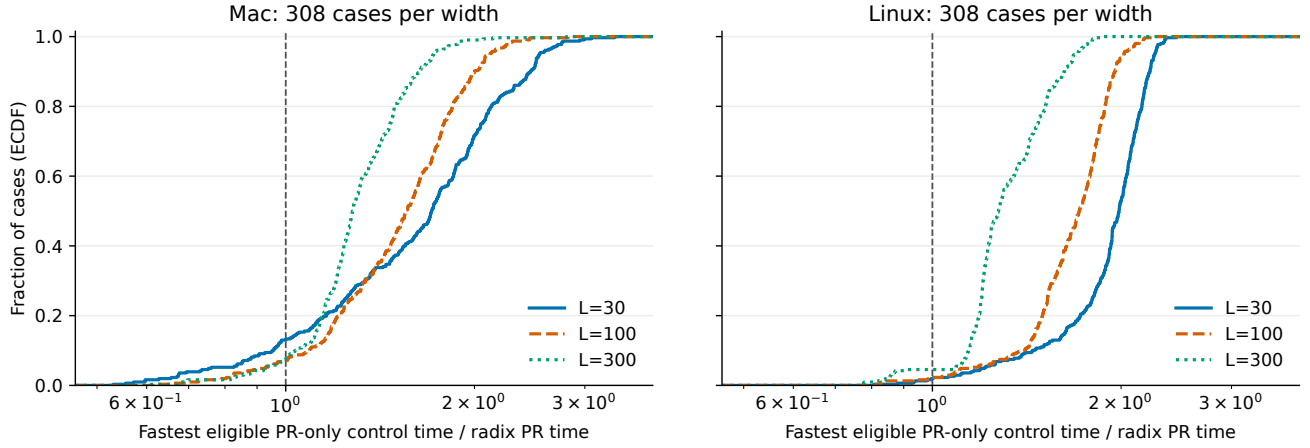


Figure 2: Post-protocol oracle ECDFs for the v3 policy holdout: all 308 cases at each width on each platform, without tail clipping. The horizontal coordinate is the fastest eligible exhaustive PR-only control's whole-task case median divided by exact sparse-radix PR time. Values above one favor radix PR. The five eligible controls are specified in the text; no oracle confidence intervals are computed.

300, the corresponding sums are Mac: 10.229, 80.733, and 479.670 seconds. Linux: 11.249, 100.601, and 221.037 seconds. These quantify operator cost across a stored-score collection; they are sums of separately timed medians, not measured batch wall times. They exclude score generation and I/O and do not establish a detector-pipeline speedup.

6.2 Directly timed stored-score collection workflow

A separate protocol measures an actual local collection operation, beyond sums of separately timed task medians. Each pass reopens 308 uncompressed NPZ files, materializes the labels and three original score arrays, makes 924 full PR calls at $L = 30$, selects all best-model ties per series, and serializes the values and decisions into one JSON file. Inputs are the same 17-family policy holdout; no detector inference occurs in this operation. The stored files total

379,354,532 bytes. Each method uses the unchanged released implementation and exhaustive thresholds. All validation and preparation inside the public calls are timed.

The whole monotonic clock runs from before the first NPZ open until the result file is closed. Imports and synthetic dtype-matched JIT warmup occur once before timing; input hashing and explicit filesystem-cache warming precede every pass. Every measured pass rereads and decodes the files; it is a warm-filesystem workflow, not a cold-disk or durable-write benchmark (no `fsync`). Three replications run serially with one numerical thread on the M1 Pro. Seeded method order was frozen before timing; its draw places dense last in all three replications, so order is randomized but not position-balanced. The active desktop was not isolated; background OCR/Python activity was recorded before execution. Agent tests, builds and other benchmarks were stopped.

Every selected-model set agrees across all nine passes for all 308 series. The maximum pairwise PR difference is 1.332×10^{-15}, below the frozen 10^{-11} tolerance; model ties use 10^{-10}. Sparse radix takes 3.187 seconds at the median, giving descriptive whole-workflow ratios of 3.545× against matched dense radix and 19.973× against Klein. Both comparisons satisfy the frozen median-time criterion. Three observations per method do not establish a confidence interval or timing significance. This directly demonstrates savings for the specified stored-score operation; it does not measure a full training/inference pipeline, production deployment, or a new-domain evaluation. An independent standard-library auditor checks recorded order, counts and clock consistency and recomputes numeric comparisons and decisions from public hash-bound receipts, without raw input arrays or a new timing measurement.

6.3 Incremental saturation: primary failure and larger-width gains

All 666 cases across Mac development, Mac external and Linux external meet the frozen 10^{-11} PR tolerance. Maximum absolute discrepancy is 1.34×10^{-15} on development and 3.67×10^{-15} on each external run, rounded upward. An independently written dense oracle also constructs ramps, components, complete global thresholds and both integrals without production geometry/integration helpers. Its 8,606 small exhaustive and adversarial cases include tied scores, widths through $2n + 1$ and noncontiguous inputs. Across 51,636 public calls, maximum discrepancies are 3.34×10^{-16} for paired output and 2.23×10^{-16} for PR-only, rounded upward. These checks concern the frozen position-prefix code; they do not certify new complete-tie code or arbitrary floating-point inputs.

The incremental primary runtime criterion fails in all three runs. At $L = 30$, median ratios against released sparse radix are 1.016, 1.026 and 1.013, with 52/126, 19/48 and 21/48 candidate losses. The criterion does not hold against every matched control. Table 6 retains all eight controls, widths and slower-case counts. Large ratios against external libraries include the support-event and ranking improvements already present in the basic evaluator; they do not establish the marginal benefit of saturation.

At the prespecified stress width $L = 300$, ratios against released radix are 1.660 on development, 1.506 on Mac external and 1.627 on Linux external. There are 125/126, 48/48 and 48/48 candidate wins. External geometric-ratio bootstrap intervals are [1.417, 2.634] and [1.527, 2.982]; aggregate-ratio intervals are [1.392, 2.472] and [1.554, 2.993]. Against the retrospective fastest of all eight controls, median ratios are only 1.136, 1.113 and 1.142, with 24/126, 9/48 and 10/48 losses. The oracle is not a deployable selector.

At width 300, aggregate saturation visits are 72.1%, 57.5% and 57.5% of full-active visits; against last-positive visits they are 73.8%, 58.3% and 58.3%. Median runtime ratios against last-positive stopping are 1.198, 1.136 and 1.152. Reduced traversal does not eliminate preparation, pointer access or component bookkeeping, and some lower-work cases remain slower. Seven development series and four external series have one original interval; interval count alone does not establish the distinct-support-score hypothesis.

A separate 16-case structural study crosses four input families with $L \in \{32, 128, 512, 2048\}$ and $n = 4L + 3$. All cases meet the frozen count predictions and agree with released radix within 1.39×10^{-17}. The distinct singleton visits $3L - 1$ positions, and the two-original separating family visits $4L$; the all-tied singleton visits all active support. These are constructed mechanism checks, not estimates of real-data prevalence or a numerical comparison to another dense oracle. The complete per-case timings and the all-tied negative control remain in the structural-study artifact.

6.4 Paired external supplement and cross-platform scope

A separately frozen descriptive supplement compares paired ROC–PR public calls at $L = 30$ on the 48 external score vectors, with three warmed complete repetitions. Actual pinned Benchopt completes all 48 cases, with median ratio 34.918× and minimum 20.079×. Released paired radix has median ratio 1.017× with 19/48 candidate losses. This is a separate workload from the whole-evaluator three-score tasks and is not pooled with the Benchopt ratios in Table 2.

DTAI [10] is prescribed for the Isolation Forest vector of each of the 12 series. Ten complete all repetitions; PSM and swan hit the 30-second parent watchdog on their first timed call and later repetitions are skipped. The completed subset's median ratio is 101.106×; the two censored cases have no fabricated duration or ratio. Native parallel/fast-math compilation is retained, with one compute thread. Completed paired outputs agree within 8.22×10^{-15}. This supplement changes neither primary decision.

The external Linux workflow regenerates the specified inputs from pinned sources. Raw data and labels match Mac hashes; 11/48 score arrays match byte for byte and 37 differ after platform-dependent score generation. All 53 downloaded artifact-file hashes were checked. This reproduces the input-generation procedure and evaluator experiment, not an identical-array replay or an independent laboratory study.

6.5 Run aggregation: structured ties and a fragmentation control

The separately frozen run study evaluates 20 constructed cases (five score families at $L = 32, 128, 512, 2048$) and every one-interval case from the previous development and external corpora: 63 and 48 cases, respectively. This is reused-data evaluation, not unseen confirmation. All four methods are PR-only and include

Table 5: Measured collection wall times in seconds. The last two columns are separately summarized phase medians, not additive whole-time estimates. All nine passes and all numeric outputs are retained.

Method	Run 1	Run 2	Run 3	Whole median	Metric median	Read median
Sparse radix PR	2.902	3.187	3.721	3.187	2.816	0.299
Dense nearest radix PR	11.456	10.944	11.298	11.298	10.605	0.413
Klein fast-VUS PR	59.041	63.647	63.749	63.647	62.493	0.927

	Mac development			Mac external			Linux external		
Control	$L = 30$	$L = 100$	$L = 300$	$L = 30$	$L = 100$	$L = 300$	$L = 30$	$L = 100$	$L = 300$
Full active scan	1.005(59)	1.057(38)	1.271(15)	1.007(22)	1.041(14)	1.216(3)	1.017(19)	1.063(9)	1.218(2)
Last-positive stop	1.011(54)	1.040(42)	1.198(21)	1.012(22)	1.059(13)	1.136(6)	1.025(19)	1.041(12)	1.152(6)
Released radix	1.016(52)	1.191(14)	1.660(1)	1.026(19)	1.151(2)	1.506(0)	1.013(21)	1.202(3)	1.627(0)
Dense nearest radix	2.602(5)	4.607(0)	8.107(0)	2.835(0)	6.250(0)	13.188(0)	3.274(0)	7.689(0)	13.526(0)
fast-VUS	20.175(0)	41.291(0)	72.435(0)	16.908(0)	42.627(0)	92.597(0)	13.174(0)	28.616(0)	57.738(0)
Sparse comparison sort	2.041(9)	2.087(6)	2.285(0)	1.783(5)	1.816(5)	1.997(4)	1.630(9)	1.678(8)	1.993(8)
Dense direct	3.286(3)	5.518(0)	10.557(0)	3.478(0)	6.401(0)	13.643(0)	3.657(0)	8.412(0)	14.453(0)
Dense direct radix	2.672(5)	4.970(0)	10.124(0)	3.008(0)	6.218(0)	13.439(0)	3.023(0)	7.705(0)	14.128(0)
Fastest eight-control oracle	0.978(84)	1.006(60)	1.136(24)	0.980(32)	1.023(19)	1.113(9)	0.984(29)	1.030(18)	1.142(10)

Table 6: Median paired runtime ratio (control divided by saturation), with the number of slower saturation cases in parentheses. Cases per width: Mac development: 126; Mac external: 48; Linux external: 48. A ratio below one favors the control. Each case uses the median of three complete public-call timings. The oracle selects the fastest of the eight measured controls separately for each case, after measurement. All cases and widths are retained. Frozen primary $L = 30$ criterion: Mac development: FAIL; Mac external: FAIL; Linux external: FAIL.

validation, ranking, geometry and, for run aggregation, the square-root table. Each case has one untimed warm call per method and five seeded randomized serial repetitions on the Apple M1 Pro with one compute thread (Python 3.12.4, NumPy 1.26.4, Numba 0.59.1).

All 131 cases satisfy the numerical and work assertions. The maximum absolute discrepancy from released radix is 5.56×10^{-16}. Every width visits at most $3P_0$ score classes. For the all-tied singleton at $L = 2048$, run aggregation inspects 2,049 runs in total, while the position scan visits 2,099,201 positions. These counts support the proved separation; they do not determine wall time.

A separate untimed verifier compares 650 random cases against direct dense evaluation (maximum absolute difference 3.34×10^{-16}), and 36 small complete integrals against independently constructed Decimal arithmetic at 80-digit precision (maximum 2.60×10^{-16}). It also checks every one of 150,000 integer prefixes against Python integers, 20,009 synthetic limb subtractions, and 6,569 per-width counter reconstructions. These are internal implementation checks, separate from benchmark cases and the local error theorem.

Table 7 reports every constructed family at the largest width. Larger ratios favor run aggregation; subunit ratios are regressions. The alternating-score control defeats compression because each score class is highly fragmented. Distinct scores also offer little aggregation benefit. These are constructed witnesses, not a sample of operational detectors.

The real-score results in Table 8 are deliberately separate. At $L = 300$, run aggregation loses to active prefix on 16 of 21 development cases and 14 of 16 external cases. It is therefore an option for compressible ties, not a replacement justified by broad real-data gains. No runtime selector or population confidence interval is inferred from these reused cases.

Table 7: Constructed $L = 2048$ cases, $n = 8195$: comparator / run-aggregation median time. All five families retained; full results at every width are in the artifact.

Scores / labels	Active prefix	Radix	Fast-VUS
All tied / singleton	8.816	29.386	739.615
Distinct / original last	0.947	32.919	765.564
Eight blocks / singleton	5.126	31.187	789.399
Alternating / singleton	0.277	1.216	41.319
All tied / interval of 8	8.867	29.487	739.147

Table 8: All reused one-interval cases. Each entry is median comparator/run ratio, followed by the number of run-aggregation losses.

Corpus	L	Active prefix	Radix	Fast-VUS
Dev. ($n_c = 21$)	30	1.010 (10)	0.984 (14)	23.606 (0)
Dev. ($n_c = 21$)	100	0.921 (15)	0.998 (11)	49.776 (0)
Dev. ($n_c = 21$)	300	0.857 (16)	1.156 (9)	58.776 (0)
Ext. ($n_c = 16$)	30	1.014 (5)	1.007 (5)	22.197 (0)
Ext. ($n_c = 16$)	100	0.996 (9)	1.006 (6)	51.389 (0)
Ext. ($n_c = 16$)	300	0.835 (14)	1.075 (0)	99.247 (0)

The protocol, source inventory, input hashes, five raw repetitions and work profiles are frozen together. The complete 131-case report includes every ratio and loss. This study changes neither the

whole-method primary PASS nor the active-prefix primary FAIL decisions.

Separate acquisition-to-timing replication. A new portable entry point downloads the pinned external data and regenerates scores before freezing inputs. Mac and a fresh Linux runner each complete 20 constructed and 48 regenerated one-interval cases, with five repetitions of four methods (1,360 timed calls per host). Both pass the original parity and work checks; maximum discrepancy from radix is 2.23×10^{-16}. Raw observations and labels match their recorded source identities. Of 48 regenerated score vectors, 48 on Mac and nine on Linux match the historical Mac bytes. These are procedural replications on previously studied data, not identical-score cross-host replays or additional unseen cases. Their separate reports retain all losses and do not replace the 131-case study.

7 LIMITATIONS AND REPRODUCIBILITY

The computational target is a pinned scalar metric, not every historical VUS variant or the complete threshold surface. Exact evaluation does not establish that this metric is the best scientific measure of anomaly quality. Support-event gains depend on sparse expanded support. Position prefixes may remain long because of component coverage or large ties; complete-tie aggregation requires one original interval and charges score-class fragmentation. At fixed small L, the preceding radix evaluator is already linear in n, so an asymptotic growing-width separation does not itself promise a default-width speedup.

The proof model fixes score keys to binary64 and treats metric arithmetic, including square roots, as unit cost. It is not an arbitrary-precision bit-complexity analysis. Integer-prefix accumulation exactly sums rounded roots and the integer test certifies a sufficient stopping condition; the local mass guarantee does not establish correct rounding of the final scalar. Measured equality is tolerance-based. No general all-input floating-point error certificate or universal CPU/GPU ranking is claimed.

The external corpus is small and its four score generators share CPU preprocessing. Hardware findings are limited to recorded environments, including workstation background activity and hosted-runner variability. The earlier whole-evaluator corpus has previously studied source families. All losses, censored cases, failed incremental criteria and negative optional racing outcomes are retained. Optional width refinement and racing are not contributions of this paper; their derivations and failed runtime decisions remain in the versioned supplementary artifact [8].

The public repository contains source, tests, proofs, independent oracles, source/input hashes, frozen protocols, raw measurements and reconstruction scripts. Frozen historical evaluators are preserved alongside new candidate code. Data are fetched from pinned public sources; raw third-party data are not relicensed or bundled, and source-specific terms are documented.

AI assistance disclosure. OpenAI Codex assisted with literature retrieval, algorithm and proof development, implementation, experiment orchestration, testing and manuscript drafting. Separately prompted agent contexts constructed oracles and reviewed claims within the same workflow. These checks are internal verification, not independent laboratory replication or external peer review. The artifact records code and measured evidence; AI tools are not listed as authors.

8 CONCLUSION

Sparse VUS evaluates the exhaustive dynamic scalar metric through support events with exact contributions for omitted thresholds. Fixed-key ranking and interval geometry give a support-sensitive baseline; width activation and component unions make further work depend on a saturated prefix. For one original interval, temporal runs aggregate complete tied classes, with an integer sufficient stopping condition and a separate local numerical bound. Whole-evaluator acceleration and incremental prefix acceleration are different empirical claims: the first passes its stated primary study, while the latter fails its default-width criterion and improves qualified larger-width regimes.

REFERENCES

[1] Benchopt contributors. 2026. Time-series foundation-model benchmark: VUS metrics. https://github.com/benchopt/benchmark_tsfm Software, audited revision 2b55ccb1, 29 September 2026.

[2] Paul Boniol, Ashwin K. Krishna, Marine Bruel, Qinghua Liu, Mingyi Huang, Themis Palpanas, Ruey S. Tsay, Aaron Elmore, Michael J. Franklin, and John Paparrizos. 2025. VUS: Effective and Efficient Accuracy Measures for Time-Series Anomaly Detection. *The VLDB Journal* 34, Article 32 (2025), 26 pages. https://doi.org/10.1007/s00778-025-00907-x

[3] Erik Demaine. 2011. Lecture 7: Linear-Time Sorting. MIT OpenCourseWare, 6.006 Introduction to Algorithms, Fall 2011. https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-fall-2011/bf7d79105762bf79bbc0925438e1468a_MIT6_006F11_lec07.pdf Accessed 2026-10-21.

[4] Tom Fawcett. 2006. An Introduction to ROC Analysis. *Pattern Recognition Letters* 27, 8 (2006), 861–874. https://doi.org/10.1016/j.patrec.2005.10.010

[5] Michael Herf. 2001. Radix Tricks. https://stereopsis.com/radix.html Accessed 2026-10-21.

[6] Tom Louis Klein. 2026. fast-VUS. https://github.com/tomlouisklein/fast-VUS Software, first commit 12 October 2026; audited revision 1bae859f, 16 October 2026.

[7] Donald E. Knuth. 2000. Dancing Links. arXiv:cs/0011047 https://arxiv.org/abs/cs/0011047

[8] Youngmin Ko. 2026. Sparse VUS, version 0.3.0. https://github.com/mandu5/sparse-vus/releases/tag/v0.3.0 Public software and working-manuscript precursor, released 21 October 2026; source revision 68d6e6de627a734880e92ecc94d3ffc9eb38a89d.

[9] Qinghua Liu and John Paparrizos. 2024. The Elephant in the Room: Towards a Reliable Time-Series Anomaly Detection Benchmark. In *Advances in Neural Information Processing Systems*, Vol. 37. 108231–108261. https://github.com/TheDatumOrg/TSB-AD

[10] ML-KULeuven. 2026. dtaianomaly. https://github.com/ML-KULeuven/dtaianomaly Software, audited revision 557b4357, 13 October 2026.

[11] Radford M. Neal. 2015. Fast Exact Summation Using Small and Large Superaccumulators. arXiv:1505.05571 [cs.NA] https://arxiv.org/abs/1505.05571 Version 1, 21 May 2015.

[12] John Paparrizos, Paul Boniol, Themis Palpanas, Ruey S. Tsay, Aaron Elmore, and Michael J. Franklin. 2022. Volume Under the Surface: A New Accuracy Evaluation Measure for Time-Series Anomaly Detection. *Proceedings of the VLDB Endowment* 15, 11 (2022), 2774–2787. https://doi.org/10.14778/3551793.3551830

[13] scikit-learn developers. 2021. Precision–Recall Curve, version 1.0.2. https://github.com/scikit-learn/scikit-learn/blob/1.0.2/sklearn/metrics/_ranking.py Full-recall truncation in precision_recall_curve; inspected 2026-10-21.

[14] Emmanouil Sylligardos. 2025. Fast or Accurate? Rethinking Time Series Anomaly Detection. In *VLDB Ph.D. Workshop*. https://www.vldb.org/2025/Workshops/VLDB-Workshops-2025/PhD/PhD25_17.pdf Section 4.

[15] Emmanouil Sylligardos. 2026. FF-VUS implementation in aVUSi. https://github.com/doanthihoaithu/aVUSi Software, audited revision facf4362, 23 June 2026.

[16] Robert E. Tarjan and Jan van Leeuwen. 1984. Worst-Case Analysis of Set Union Algorithms. *J. ACM* 31, 2 (1984), 245–281. https://doi.org/10.1145/62.2160

[17] TheDatumOrg. 2026. VUS: Official Metric Implementation. https://github.com/TheDatumOrg/VUS/blob/467bace0c103f5c2bdb4900306d82939bcac8ff9/vus/utils/metrics.py Pinned revision 467bace0; RangeAUC_volume_opt with exhaustive thresholds, inspected 2026-10-21.

[18] Xiaona Zhou, Constantin Brif, and Ismini Lourentzou. 2026. mTSBench: Benchmarking Multivariate Time Series Anomaly Detection and Model Selection at

Scale. *Transactions on Machine Learning Research* (2026). https://openreview.net/forum?id=8LfB8HD1WU Code revision 0ebf90da; dataset revision 8dea9af6.

[19] Xiaokun Zhu, Louis Carpentier, and Mathias Verbeke. 2026. When Foundation Models are One-Liners: Limitations and Future Directions for Time Series Anomaly Detection. In *International Conference on Learning Representations*. https://openreview.net/forum?id=H27kvyG4qf Released scores: KU Leuven RDR, V1, doi:10.48804/MCVI4N.